



INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Employee Management System

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About me



- **Background** : MBA in Finance
- **Motivation** : Coming from a process executive background in SLA/quality tracking and business documentation, I wanted to move into a role where I work directly with data to drive decisions.
- **Experience** : 2 years as senior process executive working on SLA/quality tracking, business documentation and advanced MS Office .
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Agenda

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Business Problem

Organizations often face challenges in managing employee records, payroll, qualifications, and leave data across multiple systems. This makes it difficult to generate accurate reports and gain insights for effective workforce management and decision-making. Manual data management increases the chances of errors, duplicate records, and delays in processing employee information. A well-designed Employee Management System using SQL helps organize data, improve accuracy, simplify reporting, and support faster business decisions



Problem Statement

Organizations require a reliable system to manage employee information accurately and efficiently. This project aims to design and implement an Employee Management System using SQL that efficiently stores, organizes, and manages employee-related data within an organization. The system maintains employee details, job roles, salary information, qualifications, leave records, and payroll data in a structured relational database. It aims to improve data accuracy, reduce manual effort, support efficient data retrieval, and generate meaningful reports to help HR and management make informed decisions.

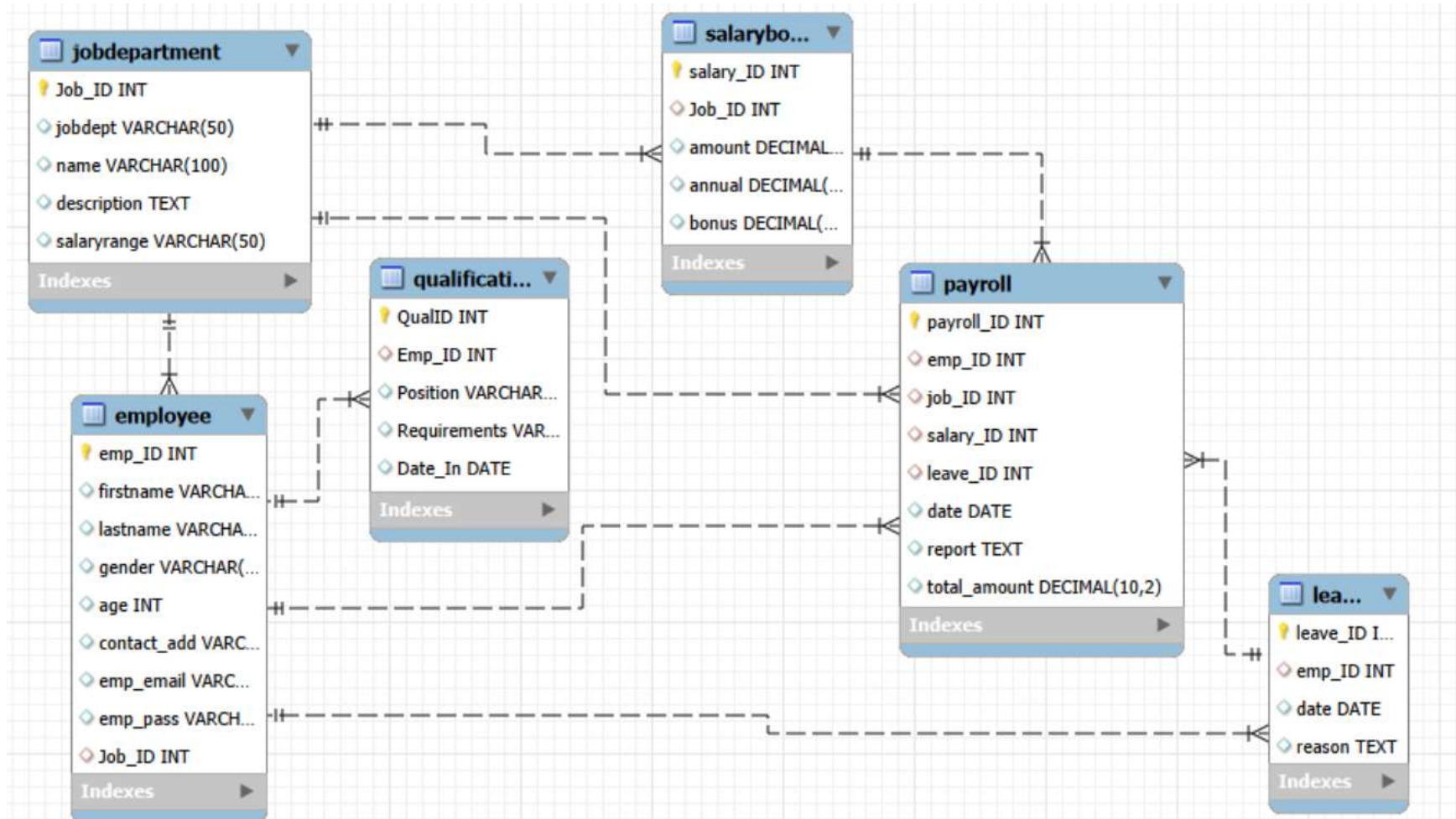


Objective of the Project

- To design and implement a centralized Employee Management System using SQL.
- To store and manage employee information efficiently in a relational database.
- To maintain data accuracy and integrity using primary keys, foreign keys, and constraints.
- To reduce data redundancy and improve consistency across employee records.
- Performed payroll, salary, qualification, and leave analysis to support HR decision-making.
- Applied SQL concepts such as JOINS, Aggregate Functions, GROUP BY, ORDER BY, and Constraints.
- To enhance decision-making by providing accurate and meaningful employee insights.

ER DIAGRAM

The database contains **six interconnected tables**, each performing a specific function.



Database

- JobDepartment Stores job roles, departments, and descriptions.
- SalaryBonus Contains salary, bonus, and annual pay for each job role.
- Employee Holds employee personal information, contact details, and job assignment.
- Qualification Maintains qualification, skills, and requirements for each employee's position.
- Leaves Tracks employee leave details such as date and reason.
- Payroll Combines employee, job, salary, and leave data to calculate final payments.
- Foreign Key Relationships The tables are connected using foreign keys with CASCADE and SET NULL rules to maintain data integrity





SQL Queries

EMPLOYEE INSIGHTS

1. How many unique employees are currently in system?

```
select count(*) as total_employees  
from employee;
```

	total_employees
▶	60

2. Which departments have the highest number of employees?

```
select jobdept, count(e.emp_ID) as employee_count  
from employee as e  
join jobdepartment as j  
on e.emp_ID=j.job_ID  
group by j.jobdept  
order by employee_count desc;
```

jobdept	employee_count
Finance	9
IT	9
Operations	8
Marketing	8
Engineering	7
Sales	7
HR	7
Legal	5

3.What is the average salary per department?

```
select j.jobdept,avg(s.amount) as avg_salary
from employee e
join jobdepartment as j
on e.job_id = j.job_id
join salarybonus as s
on j.job_id = s.job_id
group by j.jobdept;
```

jobdept	avg_salary
Operations	68750.000000
Finance	72333.333333
IT	70888.888889
Marketing	65625.000000
Engineering	81142.857143
Sales	75428.571429
HR	62571.428571
Legal	84600.000000

4. Who are the top 5 highest-paid employees?

```
select e.firstname,e.lastname,s.amount as salary
from employee e
join salarybonus as s
on e.job_id = s.job_id
order by s.amount desc
limit 5;
```

firstname	lastname	salary
Ingrid	Adams	170000.00
John	Baker	160000.00
Grake	Moor	150000.00
Hank	Wilson	150000.00
Kelly	Cooper	140000.00



JOB ROLE AND DEPARTMENT ANALYSIS

1. How many different job roles exist in each department?

```
select jobdept, count(name)
from Jobdepartment
group by jobdept;
```

jobdept	count(name)
Operations	8
Finance	9
IT	9
Marketing	8
Engineering	7
Sales	7
HR	7
Legal	5

2. What is the average salary range per department?

```
select jobdept, avg(amount) as avg_salary
from jobdepartment j
join Salarybonus s
on j.job_id = s.job_id
group by jobdept;
```

jobdept	avg_salary
Operations	68750.000000
Finance	72333.333333
IT	70888.888889
Marketing	65625.000000
Engineering	81142.857143
Sales	75428.571429
HR	62571.428571
Legal	84600.000000

3. Which job roles offer the highest salary?

```
select name, amount as highest_salary
from Jobdepartment j
join Salarybonus s
on j.job_id = s.job_id
order by amount desc
limit 1;
```

name	highest_salary
Finance Director	170000.00

4. Which departments have the highest total salary allocation?

```
select jobdept, sum(amount)
from Jobdepartment j
join Salarybonus s
on j.job_id = s.job_id
group by jobdept
order by sum(amount) desc;
```

jobdept	sum(amount)
Finance	651000.00
IT	638000.00
Engineering	568000.00
Operations	550000.00
Sales	528000.00
Marketing	525000.00
HR	438000.00
Legal	423000.00



QUALIFICATION AND SKILLS ANALYSIS

1. How many employees have at least one qualification listed?

```
select count(distinct emp_id)
from Qualification;
```

count(distinct emp_id)
60

2. Which positions require the most qualifications?

```
select position, count(requirements)
from Qualification
group by position
order by count(requirements) desc;
```

position	count(requirements)
HR Assistant	1
HR Manager	1
IT Support	1
Software Engineer	1
Data Analyst	1

3. Which employees have the highest number of qualifications?

```
select emp_id, count(*)
from Qualification
group by emp_id
order by count(*)
limit 1;
```

emp_id	count(*)
1	1



LEAVE AND ABSENCE PATTERNS

1. Which year had the most employees taking leaves?

```
select year(date), count(distinct emp_id)
from leaves
group by year(date)
order by count(*) desc;
```

year(date)	count(distinct emp_id)
2024	60

2. What is the average number of leave days taken by its employees per department?

```
select jobdept, avg(extract(day from date)) as avg_leaves
from leaves l
join employee e
on e.emp_id = l.emp_id
join jobdepartment j
on e.job_id = j.job_id
group by jobdept;
```

jobdept	avg_leaves
Operations	7.2500
Finance	13.7778
IT	10.6667
Marketing	14.2500
Engineering	15.5714
Sales	11.1429
HR	7.0000
Legal	18.6000

3.Which employees have taken the most leaves?

```
select emp_id,sum(extract(day from date)) as most_leaves
from leaves
group by emp_id
order by most_leaves desc;
```

emp_id	most_leaves
24	30
12	30
43	28
59	28
55	28
31	27
4	25
13	25

4..What is the total number of leave days taken company-wide?

```
select jobdept,sum(extract(day from date)) as total_leaves
from leaves l
join employee e
on e.emp_id = l.emp_id
join jobdepartment j
on e.job_id = j.job_id
group by jobdept
order by total_leaves desc;
```

jobdept	total_leaves
Finance	124
Marketing	114
Engineering	109
IT	96
Legal	93
Sales	78
Operations	58
HR	49



PAYROLL AND COMPENSATION ANALYSIS

1. What is the total monthly payroll processed?

```
select extract(year from date) as year,  
extract(month from date) as month,  
sum(total_amount) as total from payroll  
group by extract(year from date),extract(month from date)  
order by year,month;
```

year	month	total
2024	4	2778000.00

2. What is the average bonus given per department?

```
select jobdept,avg(bonus) as avg_bonus  
from jobdepartment j  
join Salarybonus s  
on j.job_id = s.job_id  
group by jobdept;
```

jobdept	avg_bonus
Operations	9687.500000
Finance	10666.666667
IT	10444.444444
Marketing	9125.000000
Engineering	12571.428571
Sales	11214.285714
HR	8171.428571
Legal	13300.000000

3.Which department receives the highest total bonuses?

```
select jobdept,sum(bonus) as highest_bonus
from jobdepartment j
join Salarybonus s
on j.job_id = s.job_id
group by jobdept
order by highest_bonus desc;
```

jobdept	highest_bonus
Finance	96000.00
IT	94000.00
Engineering	88000.00
Sales	78500.00
Operations	77500.00
Marketing	73000.00
Legal	66500.00
HR	57200.00

4.What is the average value of total amount after considering leave deductions?

```
select avg(total_amount) as total_amount
from payroll;
```

total_amount
46300.000000





Key Insights

- Identified highest-paid employees and total payroll expenditure.
- Analyzed department-wise salary and bonus distribution.
- Finance department recorded the highest total bonus allocation.
- Evaluated employee qualifications for workforce planning.
- Monitored leave utilization and employee attendance trends.
- Generated department-wise workforce insights.
- Improved reporting efficiency through SQL-based analysis.
- Supported better HR and management decision



Challenges

- Defining correct table relationships and ensuring accurate use of foreign keys.
- Maintaining data consistency with cascading updates and deletes.
- Writing complex joins for reports involving employee roles, leaves, and payroll.
- Ensuring all date fields follow the YYYY-MM-DD format for reliable time-based analysis.
- Preventing duplicate entries using unique constraints, especially on email fields.

Conclusion

The Employee Management System successfully demonstrates how SQL can organize and analyze large employee datasets. The database effectively stores and organizes data related to employees, departments, payroll, bonuses, qualifications, and leave records. SQL queries were used to analyze employee performance, salary distribution, bonus allocation, qualifications, and leave management. The insights generated from the system support better HR management and organizational decision-making. The project helps reduce manual effort, improve reporting efficiency, and maintain centralized employee records. Overall, the Employee Management System demonstrates how database technology can streamline workforce management and support data-driven business operations.



Q&A

THANK
YOU

